

Gabriel Castillo

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Education

Cornell University, College of Arts and Sciences

Ithaca, NY

BA in Computer Science *GPA: 3.83*

Expected May 2028

Relevant Courses: OOP & Data Structures (Honors), Discrete Math, Intro to Analysis of Algorithms, Intro to ML

Technical Skills

Languages: Python, Swift, Java, TypeScript, JavaScript, OCaml

Skills: LangChain, SQLAlchemy, PyTorch, Firebase, UIKit, SwiftUI, React Native, React, AWS, Cursor, Github Copilot

Experience

Apple

San Diego, CA

Incoming Software Engineering Intern

Expected May 2026 – Aug 2026

- o Incoming Software Engineering Intern for Summer 2026.

Cornell AppDev

Ithaca, NY

iOS Developer

Feb 2026 – Present

- o Collaborating in an 8-person Agile pod to build Navi, Cornell's primary student transit and navigation app.
- o Maintaining a 17,900+ line Swift codebase, utilizing UIKit and SnapKit for programmatic custom UI development.
- o Developing real-time TCAT bus tracking and routing features used by 5,000+ active student commuters monthly.
- o Integrating Apollo iOS to execute complex GraphQL queries, ensuring reliable live transit data from the backend.

Cornell University — uTech Lumeo (AQI-Aware Navigation App)

Ithaca, NY

iOS Developer & Researcher

Feb 2025 – Present

- o Collaborated with 4 devs to build an iOS app with Google Maps SDK and Firebase for real-time AQI-aware navigation.
- o Optimized Pareto-efficient algo to process 19,000+ AQI data points, generating routes that balance time and pollution.
- o Helped implement Google Air Quality API heatmap overlay with 16 zoom levels, visualizing real-time pollution data.
- o Extended routing logic to support 3 transport modes (drive, walk, bike), to minimize commuter pollution exposure.

rapStudy — US DoE-funded Musical Ed-Startup

Remote

Full-Stack Software Engineer Intern

May 2025 – Aug 2025

- o Refactored a 5,000+ line React codebase, improving page load times by 30% and increasing customer satisfaction.
- o Resolved major Firebase syncing issues, increasing data throughput by 40%, fixing 6 bugs in the process.
- o Delivered 3 new features that enhanced platform usability and demo success, expanding user engagement.
- o Automated testing suite and deployment system to reduce release and update cycle times by 20%.

Projects

Harbor – Cornell Connection App 

Jun 2025 – Present

- o Launched a campus-wide connection app, acquiring 40+ active users in the first week of beta testing.
- o Spearheaded full stack development with React Native (Expo), TypeScript and Firebase.
- o Built secure photo reveal system with an intermediate consent screen using GCloud Storage Signed URLs.
- o Engineered an advanced AI recommendation system that learns from a users profile and who they swipe on.

HSPorter–AI-Powered Tariff Code Classification System(YC Interview) 

Oct 2025 – Dec 2025

- o Built AI classification system achieving 75-80% accuracy across 19,000+ HS codes for import classification.
- o Implemented RAG pipeline reducing 1,264 headings to top 15 candidates, cutting LLM token costs by 98.8%.
- o Developed multi-path tree traversal algorithm with confidence-based branching for parallel classification.
- o Architected multi-LLM layer supporting 4 providers (OpenAI, Anthropic, Groq, Gemini) with hot-swappable config.

Odyssey – Interactive OCaml Theorem Prover 

Mar 2025

- o Worked with 3 others to build propositional logic theorem prover with SAT solving and tautology check.
- o Implemented recursive descent parser and AST interpreter in OCaml with 5 logical operators.
- o Developed interactive CLI with ANSITerminal for colored output and step-by-step proof visualization.
- o Added CNF conversion, DIMACS export, and LaTeX document generation for academic compatibility.

LockedIn – iOS Swipe-Based Networking App 

Dec 2024

- o Won best UI (out of 28 teams) at AppDev's Hack Challenge building a LinkedIn-Tinder hybrid native iOS app.
- o Developed frontend with Swift and UIKit; integrated Firebase for auth and real time messaging and notifications.
- o Engineered a custom card-swipe interface using gesture recognizers and animations to create a fluid user experience.
- o Helped create and design REST API with Flask, SQLAlchemy; used AWS S3 for image hosting.